



Dr. Monex Sharma

Assistant Professor | Robotics Software Engineer,
VFSTR, Guntur, Andhra Pradesh.
WCB Robotics India Pvt Ltd, Hyderabad
M. Tech & Ph.D. NIT Warangal.

Contact Information

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Websites

Links:



Profile

I completed my Ph.D. at NIT Warangal. My doctoral research focused on the design and implementation of coverage path planning algorithms for mobile robots operating on flat and uneven terrains with complex obstacles, emphasizing multi-objective optimization, energy-efficient path generation, and experimental validation. My work involved multi-objective optimization using variants of NSGA-II to improve coverage efficiency and performance. During my Ph.D., I gained hands-on experience in simulation and real-time implementation, including work on Raspberry Pi using the GoPiGo3 robot.

I am an Assistant Professor at Vignan University (VFSTR, Guntur) in the B.Tech. Robotics and Automation programme. I teach undergraduate robotics courses and conduct research in autonomous mobile robots, ROS 2, coverage path planning, optimization algorithms, and intelligent robotic systems. My technical expertise includes robot software development using C++ and Python, mobile robot navigation, and the design and implementation of robotic applications. Prior to joining academia, I worked as a Robotics Software Engineer at WCB Robotics India Pvt. Ltd., where I developed algorithms for coverage path planning, localization, SLAM, sensor fusion, control systems, and robot kinematics using C, Python, and Webots for facade cleaning robotic systems. I also worked with a facade cleaning ELMO 2.3-based robotic systems for real-world robotic applications.

Experience

06-2026
present

Assistant Professor (B. Tech in Robotics and Automation)

VFSTR (Vignan's Foundation for Science, Technology and Research), Vignan University, Vadlamudi, Guntur district.

12-2024
05-2026

Robotics Software Engineer

WCB Robotics India Pvt Ltd., Hyderabad

Link: <https://www.wcbrobotics.com/>

Education

08-2019 05-2026	Doctor of Philosophy in Mechanical Engineering. National Institute of Technology, Warangal, Telangana, India. Thesis Title: Terrain-Adaptive Multi-Objective Coverage Path Planning for Mobile Robots Using an Improved NSGA-II Algorithm.
07-2017 07-2019	Master of Engineering in Manufacturing engineering with a CGPA of 7.2. National Institute of Technology, Warangal, Telangana, India. Thesis Title: Design of a Cost-Effective, High-Strength Cartesian Robot for Pick-and-Place Applications, Integrated with Image Processing for Inventory Identification.
07-2011 05-2015	Bachelor of Engineering in Mechanical Engineering with a CGPA of 8.1 (76.69%) Shankara Institute of Technology Jaipur. Thesis Title: Trajectory Planning of Quadcopter.
2010	Class-XII , (percentage: 71) CBSE, School: Kendriya Vidyalaya Amerigog, Guwahati, Assam, India
2008	Class-X , (percentage: 72) CBSE, School: Kendriya Vidyalaya Amerigog, Guwahati, Assam, India

Courses teaching

- Mobile robotics
- Cybernetics (C++ and Python for Robotics and control system)
- ROS programming
- Inter-departmental project

Field of Specializations

- Mobile robotics
- Optimization
- Coverage path planning
- Multi-objective optimization
- Kinematics of mobile robot
- AI and sensor fusion in mobile robot
- SLAM
- Control system

List of Publications (Journal)

1. Monex Sharma, and Hari Kumar Voruganti. *Energy-Efficient Coverage Path Planning for Mobile Robots on Uneven Terrain via the TA-NSGA-II Approach*. *Arab J Sci Eng* (2025). <https://doi.org/10.1007/s13369-025-10844-0> (Q1, IF 2.7, Springer Nature, SCIE).
2. Monex Sharma, and Hari Kumar Voruganti. *Multi-objective optimization approach for coverage path planning of mobile robot*. *Robotica, Cambridge University Press* (2024). <https://doi.org/10.1017/S0263574724000377> (Q1, IF 2.7, Cambridge University Press, SCIE).
3. Monex Sharma, and Hari Kumar Voruganti. *Implementation of Coverage Path Planning Based on Modified NSGA-II and Kalman Filter on GoPiGo3 Robot*. *International Journal of Automation and Smart Technology* (2026). <https://doi.org/10.1007/s13369-025-10844-0> (Scopus).

List of Publications (Conference)

1. Monex Sharma, and Hari Kumar Voruganti. *A Classical Multi-objective Approach for Coverage Path Plans for a Mobile Robot in an Uneven Terrain*. *Advances in Multidisciplinary Design, Analysis and Optimization, NCMDAO* (2023), IIT Guwahati. Lecture Notes in Mechanical Engineering. Springer, Singapore. https://doi.org/10.1007/978-981-96-1158-4_22
2. Monex Sharma, and Hari Kumar Voruganti. *Preference Based Multi-Objective Optimization Techniques for Coverage Path Planning of a Mobile Robot*. *Proceedings of the 6th International Conference on Advances in Robotics, AIR* (2023), IIT Ropar. ACM Digital Library. <https://doi.org/10.1145/3610419.3610438>
3. Monex Sharma, and Hari Kumar Voruganti. *An Effective Genetic Algorithm Based Multi-objective Optimization Approach for Coverage Path Planning of Mobile Robot*. *Advancements in Interdisciplinary Research. AIR* (2022), NIT Allahabad. Communications in Computer and Information Science, vol 1738. Springer, Cham. https://doi.org/10.1007/978-3-031-23724-9_41

Software

Programming Languages	C, C++, MATLAB, Python
Engineering Software	Solid Works
IDE	Visual Studio Code, Spyder, Arduino, Code Composer Studio
Framework	ROS2 (Humble)

Robot Simulation Software	Webots, Gazebo
Version control system and cloud platform	GIT and GITHUB

Ongoing Research

06-2024 Present	Title: Intelligent Mobile Robot Cleaning System Under Localization Uncertainty This project develops an autonomous cleaning mobile robot for dynamic, unknown indoor environments with non-uniform dirt distribution. The robot estimates localization uncertainty and adaptively plans coverage paths to maximize cleaning efficiency, reliability, and navigation safety.
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Research Experience

12-2024 05-2026 (WCB Robotics)	Title: Localization of a façade climbing mobile robot Pose Localization of a Façade-Climbing Robot using Adaptive Kalman Filtering and Multi-Sensor Fusion (IR, IMU, Odometry, Drag Wheel, Barometer) with Signal Processing. Title: Coverage path planning of a mobile robot for a given environment based on improved NSGA-II method. Title: Slip detection using instantaneous center of rotation 4-Wheel Slip Detection using a Robust ICR Consensus Model with Adaptive Optimization for Center Point Error Minimization Title: Optimizing suction pressure for minimum energy consumption while ensuring stability and preventing detachment during robot motion and rotation. Title: Design and development of a spur gear profile with minimized backlash using Python.
05-2020 07-2022	Title: Design and development of a novel NSGA-II variant for coverage path planning of a mobile robot on flat (2D) terrain with complex obstacle geometries for warehouse applications. A proposed grid-based coverage path planning method was formulated as a multi-objective optimization problem using a novel NSGA-II variant and compared with existing methods, incorporating improved crossover and mutation operators to achieve 100% area coverage while minimizing travel distance and number of turns and enhancing obstacle avoidance in complex warehouse environments.

08-2022 10-2023	Title: Design and development of a novel terrain-adaptive NSGA-II method for coverage path planning of a mobile robot on uneven (3D) terrain with complex obstacles for agricultural and landmine detection applications. A proposed terrain-aware coverage path planning method was formulated as a multi-objective optimization problem using a novel terrain-adaptive NSGA-II approach and compared with existing methods, incorporating improved operators and a threshold pitch angle constraint for smooth motion, achieving 100% area coverage while minimizing travel distance, turns, and terrain-induced cost in uneven (3D) environments.
11-2023 12-2024	Title: Experimental validation of coverage path planning on a differential drive GoPiGo3 robot using Raspberry Pi, with adaptive Kalman filtering and multi-sensor fusion, and PD control. A proposed coverage path planning method was experimentally validated on a differential drive GoPiGo3 robot using Raspberry Pi and compared with existing methods, incorporating adaptive Kalman filtering, multi-sensor fusion, and PD control tuned using the Ziegler-Nichols method for improved localization and motion performance.

Conferences Attended

1. Advances in Robotics, 6th International Conference of The Robotics Society (**AIR 2023**), 5-8 July 2023, Department of Mechanical Engineering, Indian Institute of Technology Ropar India.
2. 6th National Conference on Multidisciplinary Design, Analysis and Optimization (**NCMDAO 2023**), 6-8 December 2023, Indian Institute of Technology Guwahati, India.
3. International Conference on Advancements in Interdisciplinary Research (**AIR 2022**), 6-7 May 2022, Department of Mechanical Engineering, Motilal Nehru National Institute of Technology (MNNIT) Allahabad, India.

Faculty Development Programs (Delivered)

1. Topic: Robotics

15-07-2024

Sanjay Gandhi Polytechnic, Ballari-583101, Karnataka, India (NBA-Accredited)

2. Topic: Smart City

12-12-2024

Sanjay Gandhi Polytechnic, Ballari-583101, Karnataka, India (NBA-Accredited)

3. Topic: Renewable Energy

14-03-2026

Sanjay Gandhi Polytechnic, Ballari-583101, Karnataka, India (NBA-Accredited)

M. Tech and B. Tech Projects Guided

1. Pradeep Lama (M. Tech) “**Slip-Aware Path Planning using Sensor Fusion**” in 2022-23, NITW.
2. Aakash Karki (M. Tech) “**Obstacle-Aware Multi-Objective Coverage Path Planning of a Mobile Robot**” in 2023-24, NITW.
3. Rohit, Subham (B. Tech) “**Implementation of coverage path planning of mobile robot using Arduino board based on odometry and distance sensor**” in 2023-24 NITW.

Courses with Certification

1. **Programming for Everybody (Getting Started with Python) - Coursera**
Instructor: Charles Russell Severance
2. **Python Data Structures - Coursera**
Instructor: Charles Russell Severance
3. **Genetic Algorithms in Python and MATLAB - Udemy**
Instructor: Yarpiz Team
4. **ROS For Beginners (ROS Noetic, Melodic, Kinetic) – Udemy**
Instructor: Edouard Renard
5. **Self-Driving and ROS - Odometry & Control – Udemy**
Instructor: Antonio Brandi

Reference

1. **Dr. Hari Kumar Voruganti**, Professor, Department of Mechanical Engineering, National Institute of Technology, Warangal, India – 506004.

Email id: harikumar@nitw.ac.in, Mobile: +91 8985929103

2. **Mr. Ujjawal Aggarwal**, Founder and Director of WCB Robotics India Pvt Ltd, Hyderabad, India 500019.

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